Answers and talk notes

Division Expressions with Fractions and Whole Numbers

What your student is learning

Tonight we divide a fraction BY a whole number — splitting a part into even smaller equal shares.

Answers

Ask your student to explain before you look.

1. **B.** 6/1. 6 = 6/1. Any whole number can be written over 1.
2. **D.** 1/6. The reciprocal of 6 is 1/6, because 6 = 6/1 and flipping gives 1/6.
3. **A.** 1/5. 2/3 ÷ 5 = 2/3 × 1/5. Dividing by 5 is the same as multiplying by its reciprocal, 1/5.
4. **C.** 2/5 × 1/3. 2/5 ÷ 3 becomes 2/5 × 1/3. Keep 2/5, change ÷ to ×, and flip 3/1 to 1/3.
5. **B.** 2/5. 4/5 ÷ 2 = 4/5 × 1/2 = 4/10 = 2/5.
6. **D.** 1/6. 2/3 ÷ 4 = 2/3 × 1/4 = 2/12 = 1/6 gallon in each jar.

Talk about it

Is one share more or less than the 3/4 you started with, and why?

Home mission

Fold a sheet of paper in half, then in half again, to make fourths. Shade 3 fourths. Pretend it is the bag of almonds. Fold each fourth in half. What fraction of the whole sheet is one cup?

When they are stuck, ask

- How many equal shares are we splitting this into?
- What is the reciprocal of that whole number?
- Should each share be bigger or smaller than what we started with?

Try not to say

- Flip the first fraction instead.
- Dividing always makes the answer bigger.
- You do not have to simplify the answer.